

Experiment No.: 03

Title: Jenkins installation



Aim: Installation of Jenkins for windows/Ubuntu

Resources needed: Internet

Theory:

Jenkins is an open-source automation server written in Java. It is used to continuously build and test software projects, enabling developers to set up a CI/CD environment. It also supports version control tools like Subversion, Git, Mercurial, and Maven. Jenkins achieves Continuous Integration with the help of plugins. Plugins allow the integration of various DevOps stages. To integrate a particular tool, you need to install the plugins for that tool. For example, Git, Maven 2 project, Amazon EC2, HTML publisher etc. It is built with Java and hence, it is portable to all the major platforms.

Jenkins features:

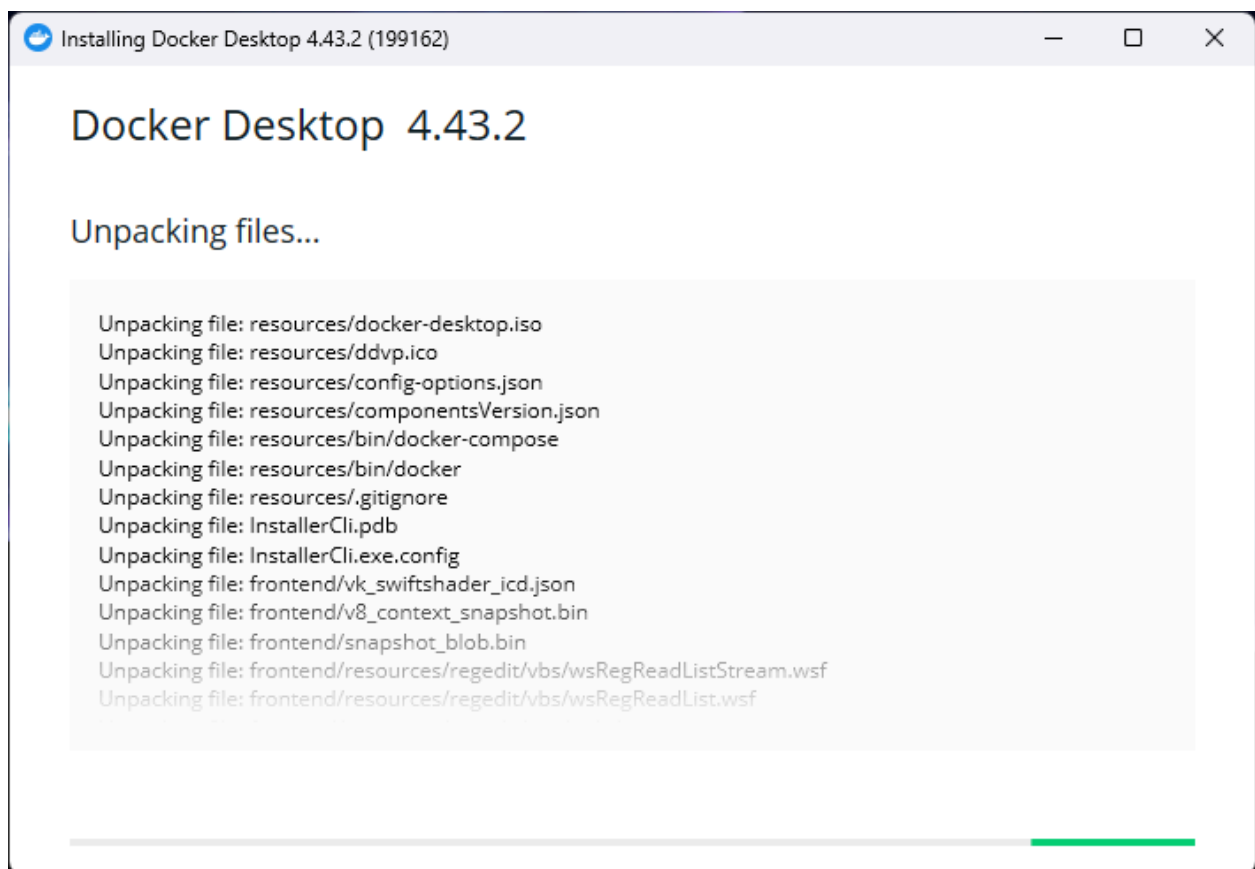
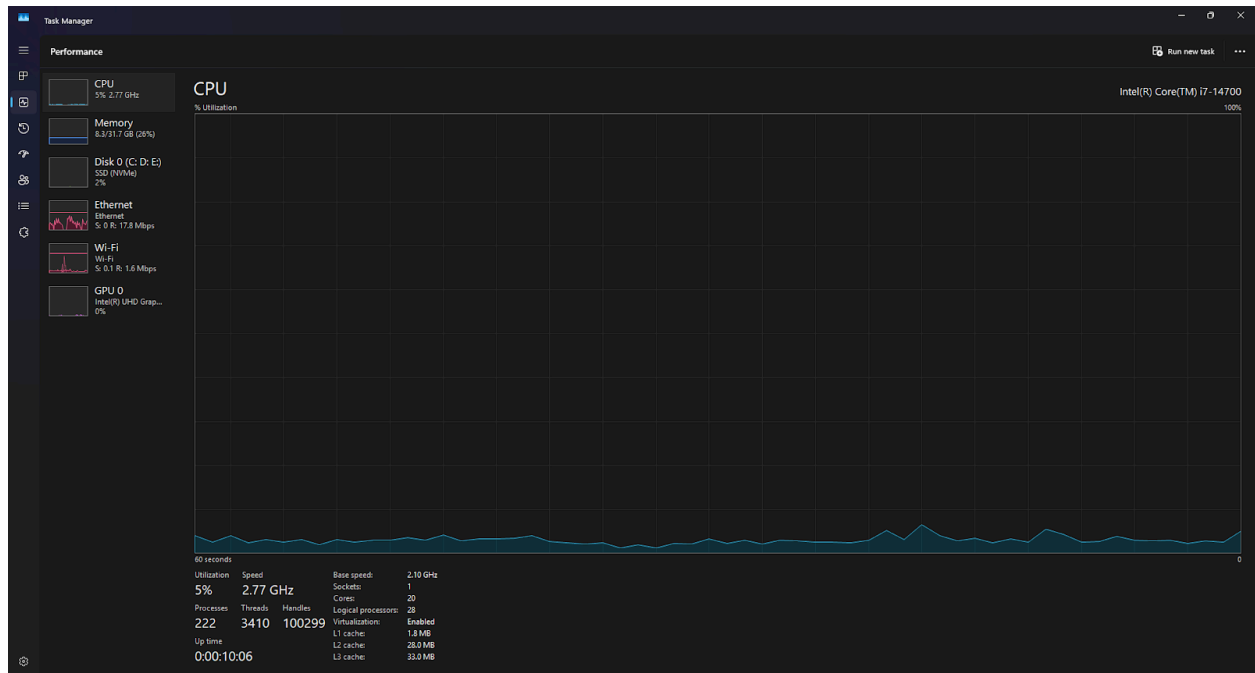
- Adoption: Jenkins is widespread, with more than 147,000 active installations and over 1 million users around the world.
 - Plugins: Jenkins is interconnected with well over 1,000 plugins that allow it to integrate with most of the development, testing and deployment tools.
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Procedure:

1. Check machine virtualization is ON.
 2. Download and install the latest version of Docker Desktop for Windows.
 3. Follow the usual installation instructions to install Docker Desktop. Depending on which version of Windows you are using, Docker Desktop may prompt you to turn on WSL 2 during installation. Read the information displayed on the screen and turn on the WSL 2 feature to continue.
 4. Start Docker Desktop from the **Windows Start** menu and minimize it.
 5. Start cmd or powerShell and Install Jenkins in docker [reference 1].
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Results: (Document with screenshots)

1. All steps of installation



The screenshot shows the Docker Desktop interface. On the left is a sidebar with navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main area is titled 'Containers' and shows a table of running containers. One container, 'admirer_rubin', is running, using the 'jenkins/jenkins:its' image. Below the table is a terminal window showing the command used to start the container: `docker run -p 8080:8080 -p 50000:50000 -d -- /var/run/docker.sock:/var/run/docker.sock -- jenkins_home=/var/jenkins_home jenkins/jenkins:its`. The bottom status bar indicates the engine is running, with RAM at 3.78 GB and CPU at 0.07%.

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
admirer_rubin	f3bd5cf616c1	jenkins/jenkins:its	8080:8080	0%	6 seconds ago	[Stop] [Restart] [Refresh]

```
PS C:\Users\EXAN_160CEB116A-01> docker run -p 8080:8080 -p 50000:50000 -d -- /var/run/docker.sock:/var/run/docker.sock -- jenkins_home=/var/jenkins_home jenkins/jenkins:its
f3bd5cf616c11532a7c2b2d3d6d9e3da7ba7f3635ac77772f8ba7c5ae4155a8a809
PS C:\Users\EXAN_160CEB116A-01>
```

The screenshot shows the Jenkins 'Unlock Jenkins' dialog box. It explains that a password has been written to the log and this file on the server: `/var/jenkins_home/secrets/initialAdminPassword`. It asks the user to copy the password from either location and paste it below. There is a text input field for the 'Administrator password' and a 'Continue' button at the bottom right.

Getting Started

Unlock Jenkins

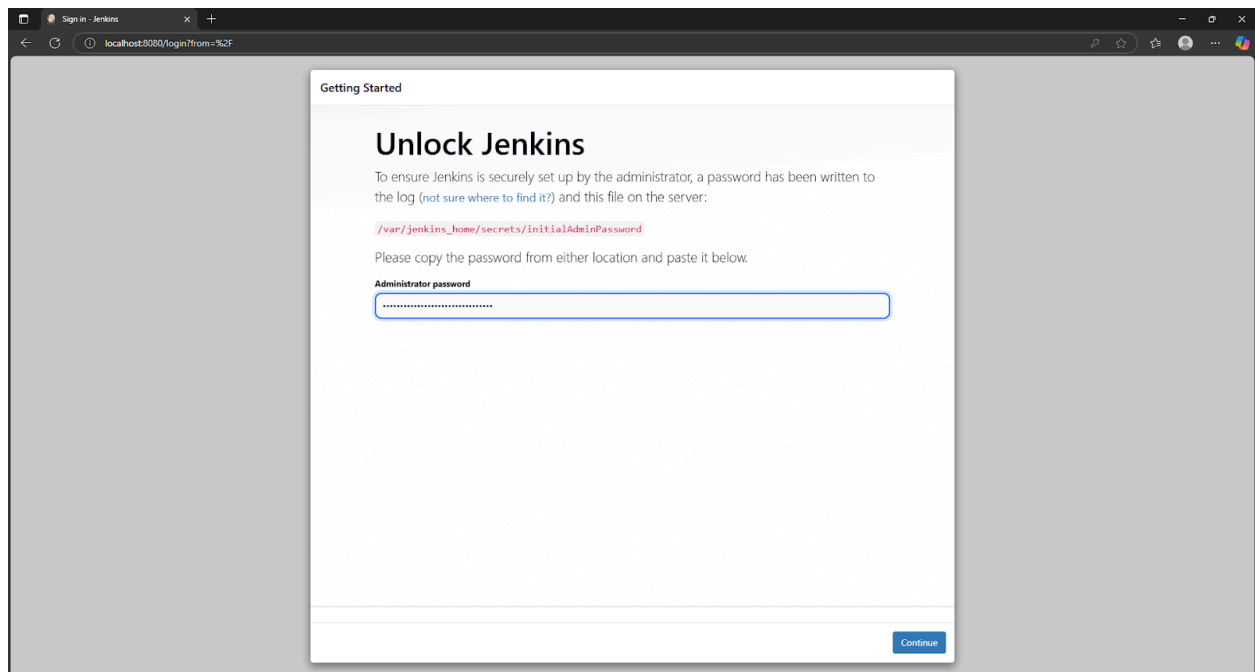
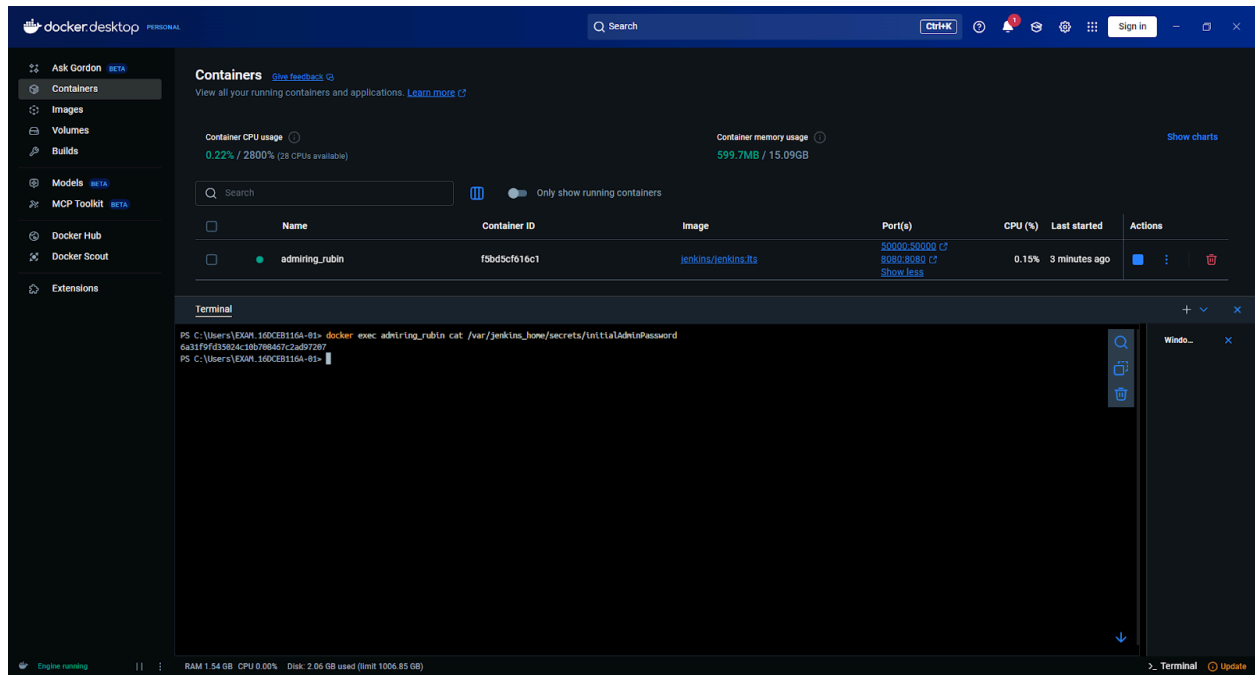
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

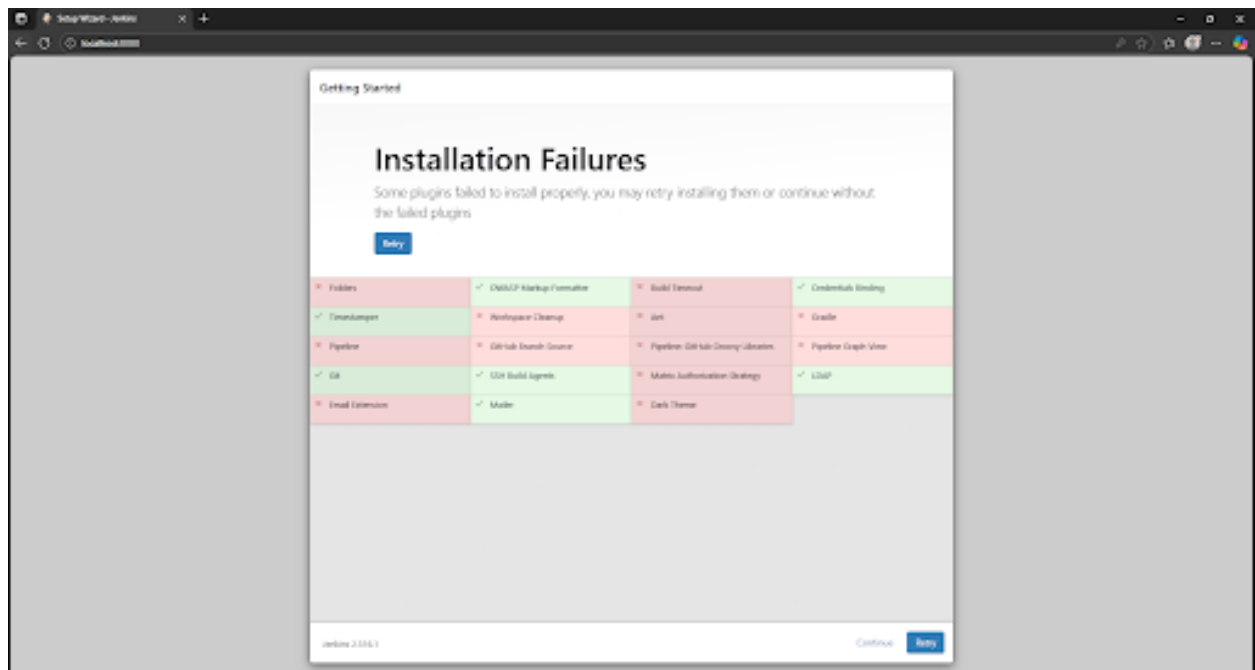
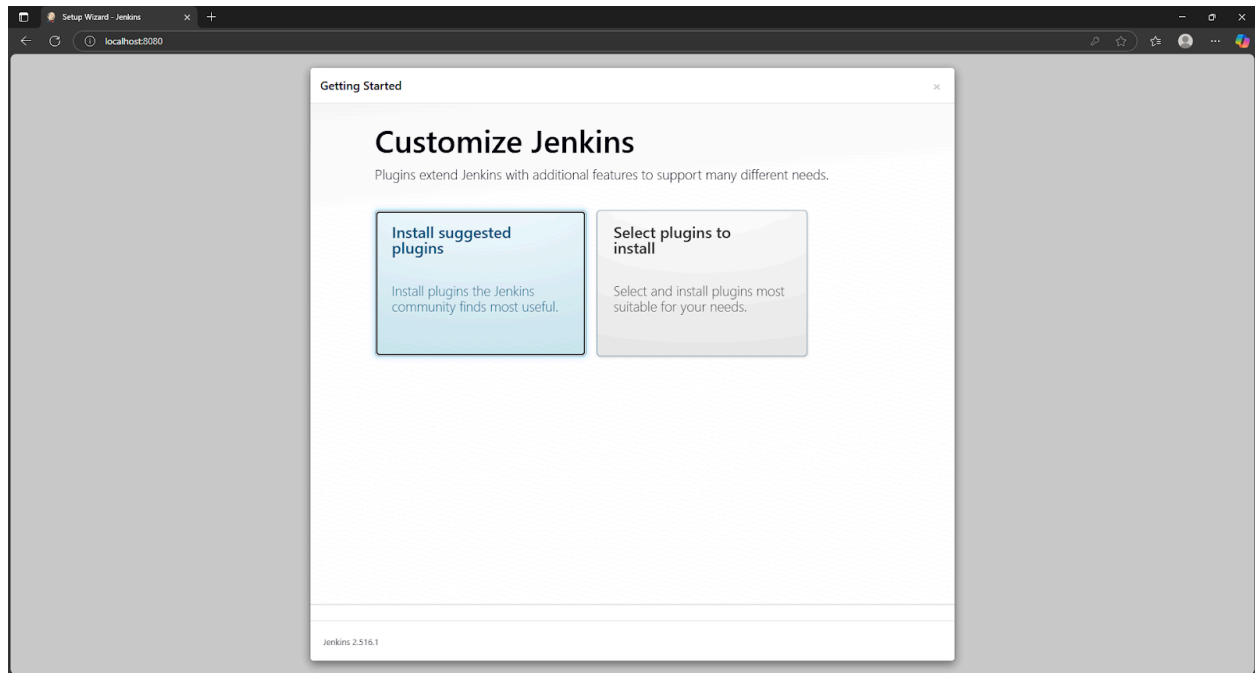
`/var/jenkins_home/secrets/initialAdminPassword`

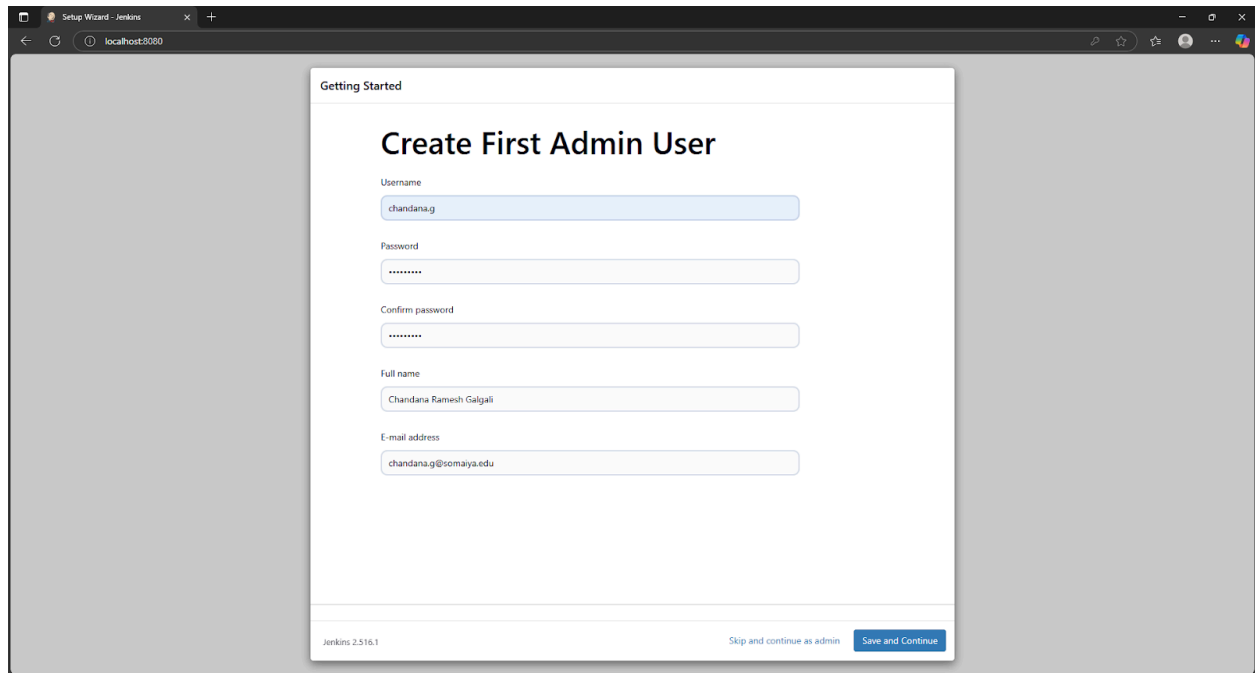
Please copy the password from either location and paste it below.

Administrator password

[Continue](#)







The image shows the Jenkins Setup Wizard 'Getting Started' screen. It is titled 'Create First Admin User'. The form contains the following fields: Username (chandana.g), Password (masked with dots), Confirm password (masked with dots), Full name (Chandana Ramesh Galgali), and E-mail address (chandana.g@somaiya.edu). At the bottom, there is a 'Skip and continue as admin' link and a 'Save and Continue' button. The version 'Jenkins 2.516.1' is displayed in the bottom left corner.

Getting Started

Create First Admin User

Username
chandana.g

Password
.....

Confirm password
.....

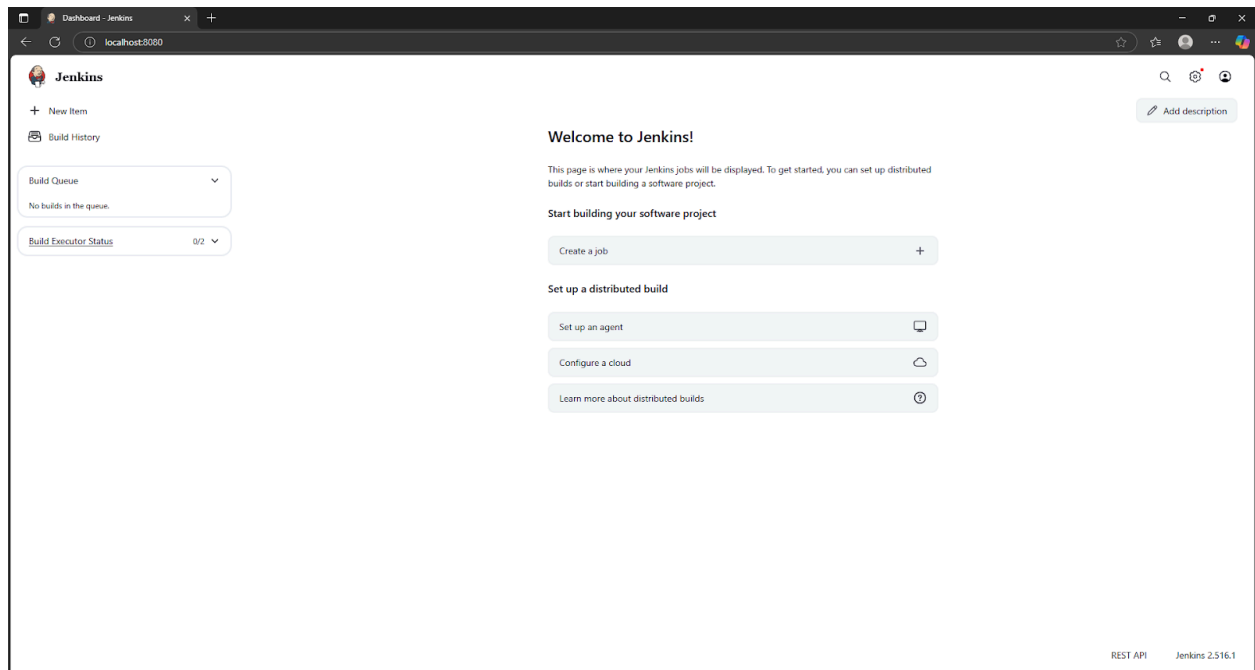
Full name
Chandana Ramesh Galgali

E-mail address
chandana.g@somaiya.edu

Jenkins 2.516.1

[Skip and continue as admin](#) [Save and Continue](#)

2. Jenkins dashboard



The image shows the Jenkins Dashboard. On the left, there is a sidebar with 'New Item' and 'Build History' links. Below these are two status boxes: 'Build Queue' (No builds in the queue) and 'Build Executor Status' (0/2). The main content area is titled 'Welcome to Jenkins!' and includes a description: 'This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.' Below this, there are three sections: 'Start building your software project' with a 'Create a job' button, 'Set up a distributed build' with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The bottom right corner shows 'REST API' and 'Jenkins 2.516.1'.

Jenkins

+ New Item

Build History

Build Queue
No builds in the queue.

Build Executor Status
0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API Jenkins 2.516.1

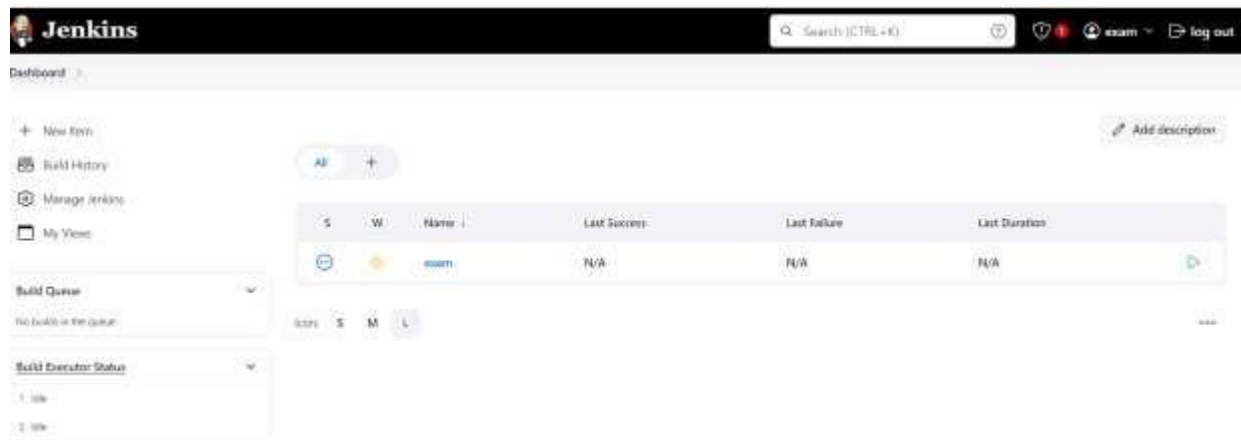


Figure 1: Jenkins Dashboard

Questions:

1. Explain various options/features of Jenkins dashboard

Ans: The Jenkins Dashboard provides the main interface for managing jobs and monitoring builds. Key features:

- New Item: Create new jobs (Freestyle, Pipeline, Multibranch Pipeline, etc.).
- People: Displays users and their activities.
- Build History: Shows a timeline of recent builds.
- Manage Jenkins: Admin section for configuring system settings, plugins, security, etc.
- My Views: Allows users to create custom views for specific jobs.
- Credentials: Manage stored credentials (API keys, SSH keys, etc.).
- Nodes: View and manage agents (nodes) for distributed builds.
- Queue: Displays jobs waiting to run.
- Log Output: For each job, logs are available to debug or audit build activity.
- Plugins: Access to install and manage over 1000 plugins to extend Jenkins functionality.

Outcomes: CO3 – Comprehend the effective code building and testing process

Conclusion: (Conclusion to be based on the Results and outcomes achieved)

In this experiment, Jenkins was successfully installed using Docker on Windows/Ubuntu. The Jenkins Dashboard was accessed and set up to manage jobs and pipelines efficiently. The experiment helped in understanding Continuous Integration and the automation of the software build and testing process. The dashboard provides a user-friendly interface to monitor build statuses, manage users, configure tools, and extend functionality with plugins.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

References:

Books/ Journals/ Websites:

1. <https://faun.pub/jenkins-in-docker-docker-running-inside-a-jenkins-container-6d02ec41b055>
 2. Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale, Jennifer Davis, Ryn Daniels, O'Reilly Media June 2016.
 3. Practical DevOps: Implement DevOps in your organization by effectively building, deploying, testing, and monitoring code,Joakim Verona, Packt Publishing, 2nd Edition,2016
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